Two-Stage Spatial-Frequency Joint Learning for Large-Factor Remote Sensing Image Super-Resolution

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***Abstract*— Super-resolution (SR) neural networks have recently achieved great progress in restoring high-quality remote sensing images (RSIs) at low zoom-in magnitudes. However, these networks often struggle with challenges like shape distortion and blurring effects due to the severe absence of structure and texture details in large-factor remote sensing image super-resolution (RSISR). Addressing these challenges, we propose a novel two-stage spatial-frequency joint learning network (TSFNet). TSFNet innovatively merges insights from both spatial and frequency domains, enabling a progressive refinement of SR results from coarse to fine. Specifically, different from existing frequency feature extraction approaches, we design a novel amplitude-guided-phase adaptive filter (AGPF) module to explicitly disentangle and sequentially recover both the global common image degradation and specific structural degradation in the frequency domain. In addition, we introduce the cross-stage feature fusion design to enhance feature representation and selectively propagate useful information from stage one to stage two. Quantitative and qualitative experimental results demonstrate that our proposed method surpasses state-of-the-art techniques in large-factor RSISR. Our code is available at https://github.com/likakakaka/TSFNet_RSISR.**

***Index Terms*— Fourier transform, large-factor remote sensing image super-resolution (RSISR), two-stage spatial-frequency joint learning (TSFNet).**

I. Introduction

REMOTE sensing image super-resolution (RSISR) is a crucial image processing technology used in various remote sensing applications where the spatial resolution is constrained by imaging equipment or poor imaging conditions [1], [2]. By recovering high-resolution (HR) remote sensing images (RSIs) from low-resolution (LR) ones, RSISR enables the extraction of more detailed information and improves the accuracy of subsequent analysis and decision-making processes.

Notably, large-factor RSISR, entailing scaling factors such as $\times$ 8 or $\times$ 16, as investigated by Meng et al. [3] and Wang et al. [4], confronts the big challenge of mitigating the significant degradation that deeply affects the quality of RSIs. Such degradation may stem from factors such as atmospheric interference, sensor limitations, and constraints inherent to imaging platforms [5], [6]. The exploration of large-factor RSISR through advanced algorithms and techniques holds remarkable importance due to its potential to transcend the boundaries imposed by LR sensors, enabling the capture of finer details that were previously unattainable. Moreover, it broadens the application boundaries of state-of-the-art super-resolution (SR) techniques and significantly augments the interpretation and analysis of extensive large-scale remote sensing scenarios. The implications of these advancements are far-reaching, particularly in fields such as aerial reconnaissance, disaster response, and satellite imaging [3], [4], [7].

With the rapid development of deep convolutional neural networks (CNNs) and transformer-based networks, numerous studies [8], [9], [10] are proposed for the RSISR task. Due to their inherent capability to capture local details or global similarities in the spatial domain [5], these approaches have demonstrated superior performance compared to traditional methods [7], particularly for low zoom-in magnitudes in RSISR [3].

Nevertheless, under large-factor degradation scenarios, where the spatial resolution of satellite images is significantly lower, substantial loss of high-frequency information will result in severe shape distortion and blurring effects [3], [6]. For instance, as illustrated in the first row of Fig. 1, these distortions are particularly evident in the wings and tails of aircraft, thus complicating feature extraction and accurate shape recovery. Furthermore, satellite images often have wide coverage, encompassing a diverse range of ground objects at various scales [11]. This diversity, coupled with the absence of specific prior knowledge and customized solutions, poses unique challenges. CNN-based methods, such as DCM [12], often yield blurry results in large-factor SR due to their limited

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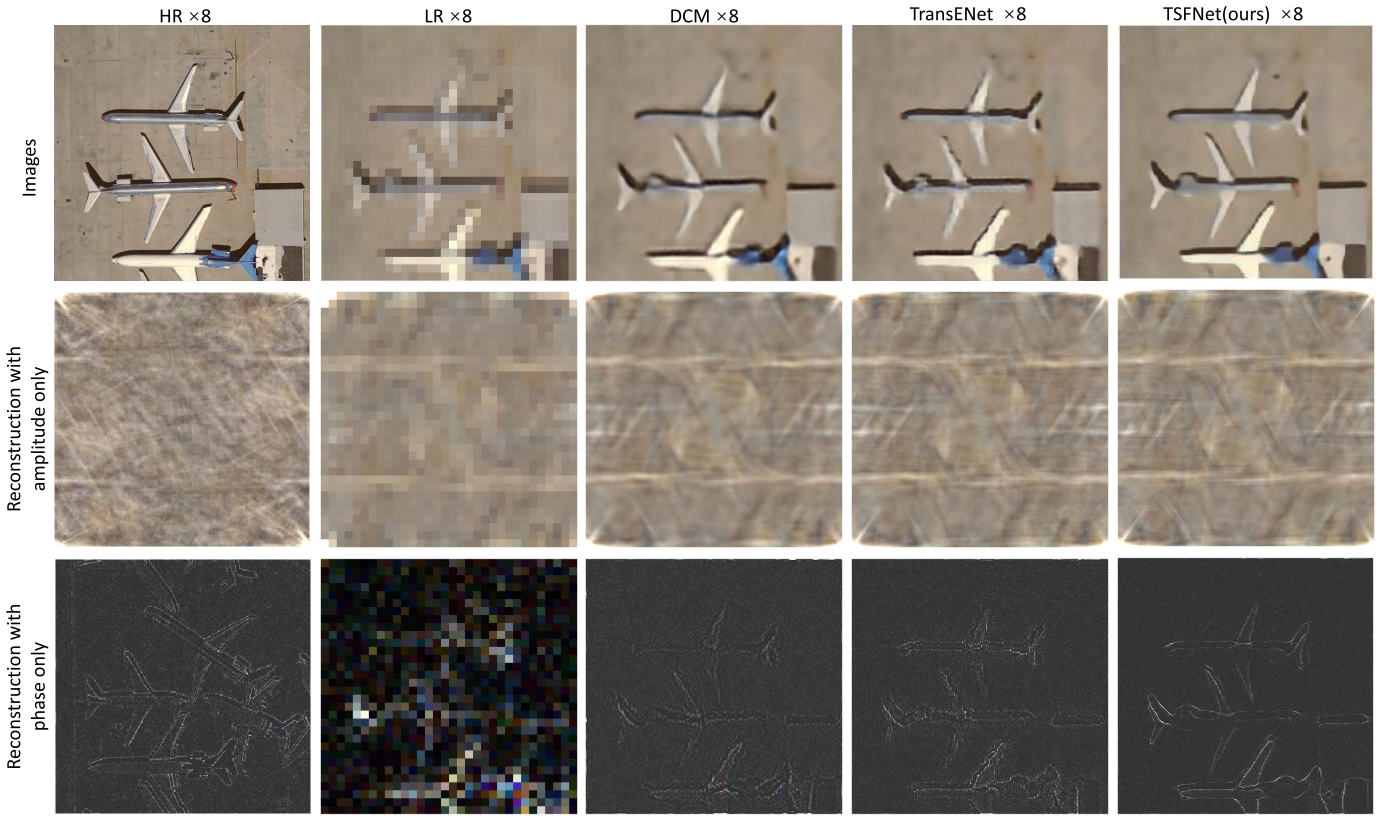


Fig. 1. Visualizing the characteristics of the amplitude spectrum and phase spectrum in the frequency domain through reconstruction. Compared with the original images (first row), we observe that the reconstructed images using only the amplitude information (second row) show that the amplitude information reflects more common image properties such as lighting, brightness, and color information. On the other hand, the reconstructed images using only the phase information (third row) indicate that the phase information reflects more specific structural degradation such as contour and critical scattering points. In comparison to other methods, although our algorithm achieves similar performance in amplitude information recovery, our TSFNet generates superior visual results with sharper edges and contours, benefiting from its improved phase information recovery.

receptive fields and local information distortion. Transformer-based methods, like transformer-based enhancement network (TransENet) [13], primarily focus on extracting global self-similarity information. This approach, while useful in some contexts, can lead to the overemphasis of similar distorted features, resulting in edge distortions in large-factor RSISR.

The amplitude and phase separation in Fourier transformation is capable of disentangling common image degradation such as lighting, brightness, and color, as well as specific structural degradation such as contour and critical scattering point, to some extent [14], [15]. Meanwhile, the process of Fourier transform inherently captures global properties, where each pixel in the Fourier space is associated with all spatial pixels under the same frequency components, thereby aiding in the recovery of details pertaining to objects sharing similar frequency bands across varying scales. These meaningful physical characteristics can be effectively utilized in the frequency feature extraction to complement the local features extracted in the spatial domain [15]. Therefore, collaboratively leveraging the combined strengths of spatial and frequency domain features is hopeful to enhance the ability to accurately recovering missing structural and textural details. Recently, some researchers have begun to introduce the idea of frequency feature extraction by Fourier transform into SR task [16], [17], [18], [19]. Nevertheless, they mainly concentrate on the global property of Fourier transform and just utilize Fourier convolutions [20] to process complex frequency domain features. They do not exploit or design dedicated modules to fully harness the disentanglement potential of Fourier transform through amplitude and phase separation.

In this article, we address these limitations and introduce a novel two-stage spatial-frequency joint learning network (TSFNet) specifically tailored for large-factor RSISR. Our approach, distinct from methods that primarily extract features from the spatial domain [8], [9], [13], incorporates a novel amplitude-guided-phase adaptively filter (AGPF) block which leverages Fourier transform to delve into the characteristics of frequency domain features. Considering the disentanglement property of the Fourier transform between common image degradation and specific structural degradation, we separate the phase information from the amplitude after the Fourier transform. In scenarios of large-factor RSISR, where structural degradation is more severe, we employ the refined amplitude residual information as guidance to reconstruct the more intricate phase information. Furthermore, we propose a two-stage progressive feature learning architecture to divide the overall recovery process into two manageable steps that progressively improve the SR quality from coarse to fine. Each stage employs an encoder–decoder design, supplemented by a novel cross-stage feature selection (CFS) module. This module allows the network to selectively enhance and propagate knowledge from stage one to stage two, thereby refining the lost details and boosting performance in the reconstruction. Our TSFNet demonstrates state-of-the-art performance in

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large-factor RSISR on three publicly available remote sensing datasets.

In summary, this work makes three main contributions.

1) We propose TSFNet for large-factor RSISR. Our network effectively extracts the dual-branch spatial and frequency information from coarse to fine, achieving competitive large-factor RSISR performance on three public remote sensing datasets.
2) To enhance the restoration of global-specific structural features in the frequency domain, we incorporate the Fourier-based disentanglement prior into the AGPF block. This integration enables the refinement of structural information with the guidance of the common recovery information, leading to more effective recovery of degraded structural information in LR RSIs.
3) We introduce the CFS module to selectively enhance and propagate knowledge from stage one to stage two. This module plays a crucial role in improving the overall performance of the network by facilitating the fusion of features across stages.

The remaining sections of this article are structured as follows. Section II provides an overview of related works in the field of image SR. In Section III, we introduce the framework of our proposed TSFNet and provide a comprehensive description of its model architecture. Section IV presents the detailed experimental setting and Section IV presents the detailed experimental settings and analyzes the quantitative and qualitative results. Lastly, Section V presents the conclusions.

II. Related Work

A. Deep Learning in Spatial Domain for Image SR

In recent years, deep learning has made significant progress across various fields [21], [22], [23], [24]. Its application in the natural image SR primarily concentrates on the spatial domain and has extensively explored diverse techniques to enhance the resolution of LR images. These methods have achieved notable improvements in visual quality and the preservation of finer details.

A key development in this area is the application of CNNs. Dong et al. [25] pioneering CNN-based method, SR convolutional neural network (SRCNN), marked a breakthrough in image SR by learning complex mappings between LR and HR image patches. Since then, numerous variations and improvements have been proposed, such as very deep convolutional network (VDSR) [26], enhanced deep super-resolution network (EDSR) [27], residual dense network (RDN) [28], and dense deep back-projection network (D-DBPN) [29]. Another significant trend is the integration of attention mechanisms, such as in residual channel attention network (RCAN) [30], residual feature aggregation network (RFANet) [31], image restoration using swin transformer (SwinIR) [32], and hybrid attention transformer network (HATNet) [33]. Attention mechanisms allow the network to focus on the global relevant image regions, enabling more accurate reconstruction of high-frequency details, which have demonstrated superior performance in terms of both quantitative metrics and visual quality. Moreover, the utilization of generative adversarial networks (GANs) has also emerged as a significant trend in SR. GAN-based methods like generative adversarial network for image superresolution (SRGAN) [34], enhanced super-resolution generative adversarial network (ESRGAN) [35], structure-preserving super resolution network (SPSR) [36], and best-buddy generative adversarial network (Beby-GAN) [37] have generated more realistic image texture by incorporating adversarial loss functions and perceptual loss functions, further pushing the boundaries of this field.

B. Deep Learning-Based Methods for RSISR

Recent advancements in deep learning for natural image SR have also inspired similar exploration in RSISR [7]. In CNN-based RSISR, various approaches have emerged. Lei and Shi [38] introduced LGCNet, leveraging the "multifork" structure to reconstruct residuals between LR and HR image pairs. Lu et al. [39] proposed a multiscale residual network for precise high-frequency information reconstruction in RSISR. Pan et al. [40] implemented two types of downprojection modules within a residual backprojection block, facilitating residual learning in both global and local contexts. Furthermore, Dong et al. [41] developed a second-order learning mechanism, optimizing the utilization of multiscale information through dual feature reuse at local and global scales.

Attention-based methods have also gained prominence in RSISR. Zhang et al. [42] designed high-order attention (HOA) modules to exploit hierarchical features comprehensively. Lei et al. [8] utilized nonlocal attention to exploit hybrid-scale self-similarity information. In addition, Lei et al. [13] introduced TransENet, a transformer-based enhancement network for RSISR. Wang et al. [5] introduced a hybrid convolution-based attention mechanism to explore multiscale features in RSIs. Lu et al. [43] proposed cross-spatial pixel integration and cross-stage feature fusion-based transformer network to enhance context cognition and understanding of the RSIs.

GAN-based methods have found applications in RSISR as well. Wu et al. [44] used the saliency-guided feedback GAN to discriminate different regions and reconstruct HR RSIs. Jia et al. [2] proposed a multiattention GAN (MA-GAN) framework for RSISR. Meng et al. [3] introduced spectra-guided generative adversarial network (SpecGAN), leveraging hyperspectral images (HSIs) in GANs for large-factor optical RSIs. However, the adversarial training process in GANs often poses challenges, leading to issues such as mode collapse or oscillation. Besides, the recovered images often exhibit undesired structural distortions, and in the case of large-factor SR, GAN-generated fake errors may even alter the semantic meaning of the image.

C. Fourier Transform for Image SR

The Fourier transform is a widely employed technique for analyzing the frequency information in signals. It offers valuable global statistical content about signals and has the ability to capture long-range dependencies [15].

Recently, Fourier-domain methods have been introduced in the field of image SR. Sinha et al. [16] and Zhang et al. [17] have, respectively, introduced networks leveraging nonlocal fast Fourier convolutions [20] to enhance SR by expanding the receptive field and capturing long-range dependencies. Similarly, Chen et al. [18] introduced SwinFSR for enhancing

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the resolution of stereo images. Wu et al. [45] introduced the frequency-domain convolution branch and spectral loss based on the Fourier transform to process frequency content and mitigate the disadvantage of the spatial CNNs in capturing high-frequency content. Wang et al. [15] developed a two-branch spatial-frequency mutual network (SFMNet) to boost face SR performance. Liu et al. [46] introduced a novel training scheme called spectral uncertainty-based decoupled frequency (SUDF) for perceptual SR. Yu et al. [47] proposed a scale-aware frequency attention network that leverages the discrete cosine transform to dynamically select and retain appropriate high-frequency texture details for continuous arbitrary scale SR tasks.

Overall, these studies underscore the effectiveness of Fourier-domain methods in enhancing image details and resolution. However, most researchers only utilize Fourier convolutions [20] to process complex frequency domain features, disregarding the potential of Fourier transform to disentangle the global common image and specific structure degradation through amplitude and phase separation. In addition, when it comes to spatial and frequency domain information fusion, most existing research relies on direct concatenation or addition, failing to effectively leverage the complementary advantages of the information in two domains.

In light of these findings and limitations, our research delves deeper into the characteristics of the Fourier transform and its application in large-factor RSISR.

III. Methodology

In this section, the proposed TSFNet is described in detail. We begin by presenting an overview of the overall framework of TSFNet. Subsequently, we mainly introduce two important designs in our TSFNet: the dual-branch spatial and frequency information extraction (SFE) design and the cross-stage feature fusion design.

A. Overview of TSFNet

As shown in Fig. 2, our proposed TSFNet is a novel two-stage progressive learning network for large-factor RSISR. Each stage of our architecture follows an encoder–decoder U-Net symmetrical hierarchical design, mainly consisting of the dual-branch SFE module and multi-Dconv transposed attention (MDTA) layer [48]. The SFE module is used to extract features from both the frequency and spatial domains, while the MDTA layer is employed to fuse these two different types of features. Furthermore, when passing the first-stage features into the second stage, we utilize a Gaussian split block (GSB) [49] as a shortcut connection to alleviate the potential loss of low-frequency information in deeper features. In addition, we introduce a novel CFS module to enable the network to selectively enrich the features of the two stages and enhance SR performance.

Specifically, given a LR image I_{LR} as the input, we first upscale it to the desired size $I_{\mathrm{LR}}^{\uparrow} \in \mathbb{R}^{H\times W\times 3}$ using bilinear interpolation, which is the preprocessing we perform to reduce the excessive loss caused by extreme degradation. Subsequently, we apply a 3×3 convolution layer and an MDTA layer [48] to extract the initial shallow features $X_0 \in \mathbb{R}^{H\times W\times 3}$. These extracted features are then passed through an encoder–decoder architecture with six inverse pixelshuffle downsamplings [50] and pixelshuffle upsamplings [51]. In addition, to fully exploit and fuse the restored local detailed spatial features and global structural frequency features, we introduce the SFE module and MDTA layer at every level of the U-Net in the first stage. The final extracted decoder features $X_N \in \mathbb{R}^{H\times W\times 3}$ are reconstructed through a 3×3 convolutional layer, and we use the preprocessing bilinear interpolation image as a summation part to obtain the final stage one SR results I_{SR1}.

For the second stage, the network takes the coarse prediction I_{SR1} as inputs and predicts the final refined SR results. Differently, from the design of stage one, we incorporate an additional GSB and four CFS modules to selectively enrich and adaptively propagate the features from stage one. The remaining designs are consistent with those of the stage one. Finally, we obtain the refined SR results I_{SR2}.

We train TSFNet by using the $L1$ loss function as

$$L(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left(\|I_{\mathrm{HR}}^{(i)} - I_{\mathrm{SR1}}^{(i)}\|_1 + \|I_{\mathrm{HR}}^{(i)} - I_{\mathrm{SR2}}^{(i)}\|_1\right) \quad (1)$$

where θ represents the model parameters, and N is the number of training images.

B. Dual-Branch SFE

As shown in Fig. 3, our SFE is comprised of two distinct branches: a spatial branch and a frequency branch. The spatial branch processes half of the input feature channels through two residual-based spatial blocks (RSPBs), focusing on local spatial receptive fields. Conversely, the frequency branch processes the remaining channels through two AGPF blocks, operating in the Fourier frequency domain. This dual-branch structure enables the capture of complementary information across different receptive fields and domains. Subsequently, the information from the two branches is concatenated at the end, and an MDTA layer [48] is employed to effectively integrate these two different types of features.

Formally, given the input features $X \in \mathbb{R}^{H\times W\times C}$, we divide X along the channel dimension into two halves, namely $X = \{X_s, X_f\}$. The spatial branch, $X_s \in \mathbb{R}^{H\times W\times C/2}$, undergoes two traditional cascaded residual blocks and one 3×3 convolutional layer, enhancing detailed spatial information. The transformation in the spatial branch is represented as

$$X_s' = f_{\mathrm{RSPB}}(X_s). \quad (2)$$

In the frequency branch, $X_f \in \mathbb{R}^{H\times W\times C/2}$ is transformed into the Fourier space using Fourier transform $\mathcal{F}$, producing complex components $\mathcal{F}(X_f)$, which is expressed as

$$\mathcal{F}(X_f)(u,v) = \frac{1}{\sqrt{\mathrm{HW}}}\sum_{h=0}^{\mathrm{H}-1}\sum_{w=0}^{\mathrm{W}-1} X_f(h,w)e^{-j2\pi\left(\frac{h}{\mathrm{H}}u+\frac{w}{\mathrm{W}}v\right)} \quad (3)$$

where u and v and the coordinates of the Fourier space.

The amplitude component $\mathcal{A}(X_f)$ and phase component $\mathcal{P}(X_f)$ in the $\mathcal{F}(X_f)$ can be expressed as

$$\mathcal{A}(X_f)(u,v) = \sqrt{\mathcal{R}^2(X_f)(u,v) + \mathcal{I}^2(X_f)(u,v)} \quad (4)$$

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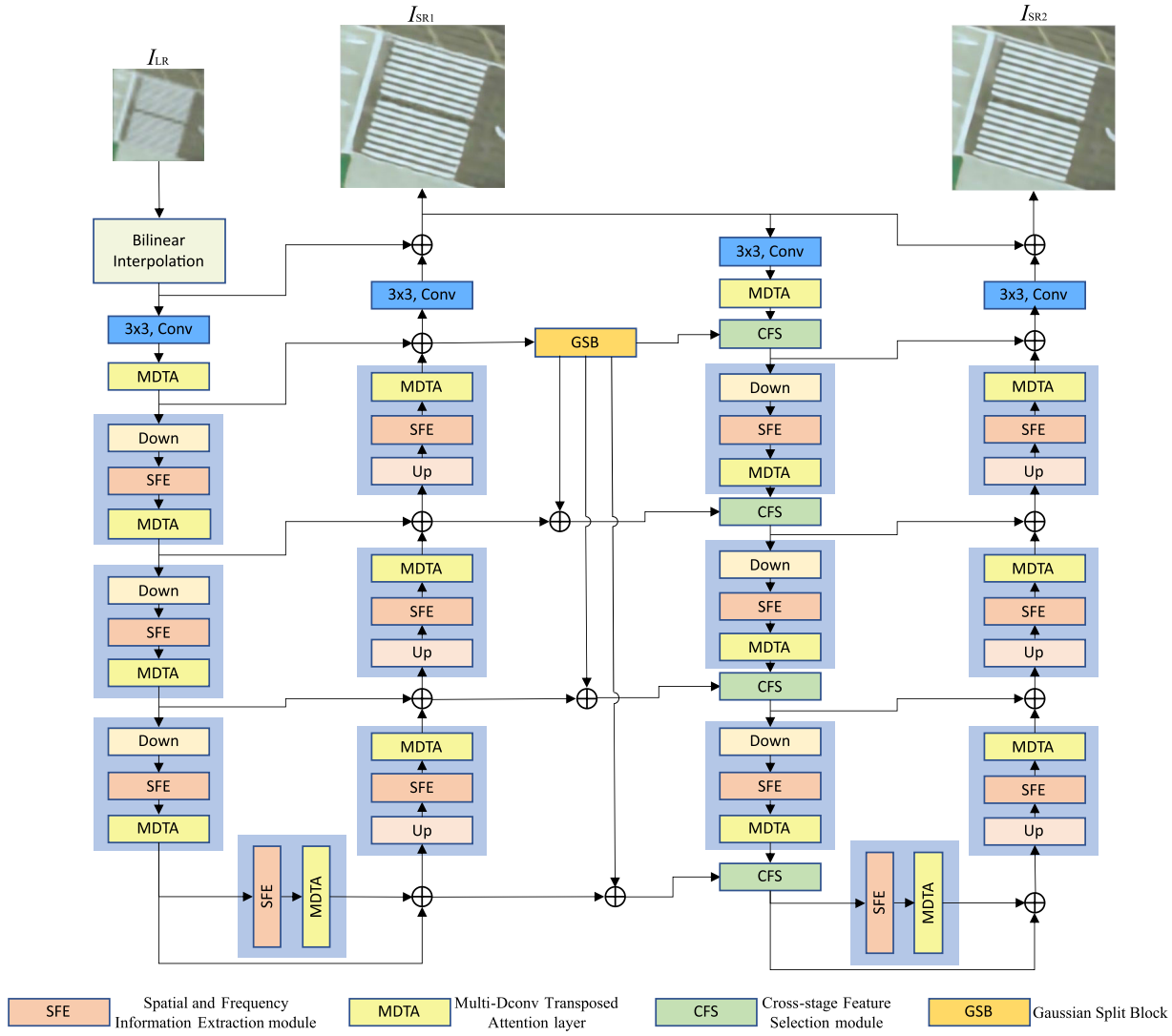


Fig. 2. Overall architecture of the proposed TSFNet. This network consists of the dual-branch SFE module, MDTA layer, GSB, and CFS module.

$$\mathcal{P}(X_f)(u,v) = \arctan\left[\frac{\mathcal{I}(X_f)(u,v)}{\mathcal{R}(X_f)(u,v)}\right] \tag{5}$$

where $\mathcal{R}(X_f)(u,v)$ and $\mathcal{I}(X_f)(u,v)$ represent the real and imaginary parts, respectively.

Considering that the amplitude component encompasses more global common image degradation, it can serve as a useful guide for restoring the more intricate phase information, which contains more specific structural degradation. We initially restore the amplitude and utilize the residual amplitude information to aid in the recovery process of the phase. Regarding the restoration of amplitude information, we first employ a 3×3 depthwise convolution $\mathrm{DW}(\cdot)$ to customize the Fourier channelwise spatial context information interaction, followed by a 1×1 point convolution $\mathrm{PW}(\cdot)$ to aggregate the Fourier pixelwise cross-channel evolution, which is expressed as

$$\mathcal{A}'(X_f) = \sigma_{a2} \cdot \mathrm{PW}(\sigma_{a1} \cdot \mathrm{DW}(\mathcal{A}(\mathrm{X_f}))) \tag{6}$$

where σ_{a1} and σ_{a2} indicate the two activation functions used, here they are both ReLU functions.

To ensure the discrepancy of the transform on various spectral locations, we introduce the generation of the frequency filters R. Initially, we extract the channel relationship on the same frequency components by applying average and maximum pooling [denoted as $P_{\mathrm{avg}}(\cdot)$ and $P_{\mathrm{max}}(\cdot)$] along the channel dimension, expressed as

$$\begin{aligned} \mathcal{A}_{\mathrm{avg}} &= P_{\mathrm{avg}}(\mathcal{A}'(X_f)) \\ \mathcal{A}_{\mathrm{max}} &= P_{\mathrm{max}}(\mathcal{A}'(X_f)). \end{aligned} \tag{7}$$

Subsequently, we concatenate the two types of pooled features and employ the same design as in (6) to transform the pooled features into the final attention map, serving as the frequency filters R, which is expressed as

$$R_a = \sigma_{a4} \cdot \mathrm{PW}(\sigma_{a3} \cdot \mathrm{DW}([\mathcal{A}_{\mathrm{avg}}; \mathcal{A}_{\mathrm{max}}]))] \tag{8}$$

where we choose σ_{a3} as the ReLU function, and σ_{a4} as Sigmoid function.

Next, we use elementwise multiplication between the frequency domain filter and the learned amplitude filter map R_a to get the final refined amplitude information $\mathcal{A}''(X_f)$, which

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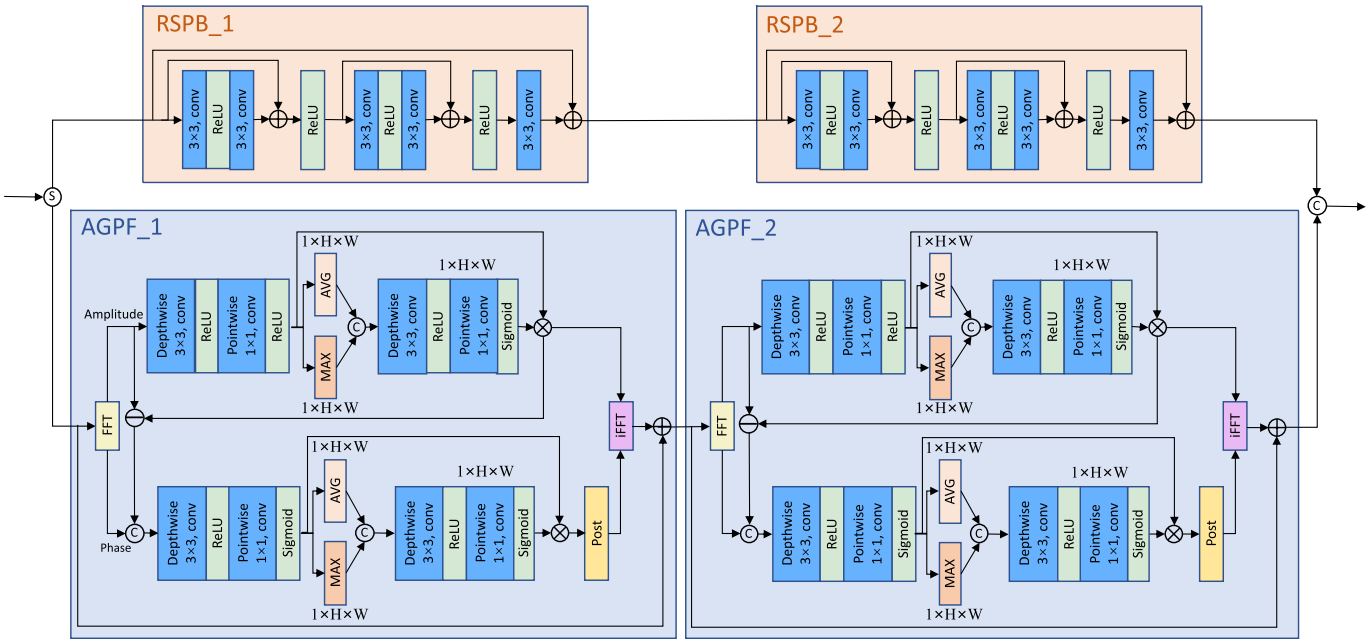


Fig. 3. Illustration of the dual-branch SFE. AGPF block designed for the frequency branch in SFE.

is expressed as

$$\mathcal{A}''(X_f) = \mathcal{A}'(X_f) \cdot R_a. \tag{9}$$

As for the restoration of phase information under the guidance of amplitude residual information, we simply concatenate the residual amplitude and phase as the input of the phase restoration network. Then, we choose the same steps as the recovery process for the above amplitude information, which is expressed as

$$\begin{aligned}
\mathcal{P}'(X_f) &= \sigma_{p2} \cdot \mathrm{PW}\big(\sigma_{p1} \cdot \mathrm{DW}\big(\big[\mathcal{P}(X_f); \mathcal{A}_{\mathrm{Res}}\big]\big)\big) \\
\mathcal{P}_{\mathrm{avg}} &= P_{\mathrm{avg}}\big(\mathcal{P}'(X_f)\big) \\
\mathcal{P}_{\max} &= P_{\max}\big(\mathcal{P}'(X_f)\big) \\
R_p &= \sigma_{p4} \cdot \mathrm{PW}\big(\sigma_{p3} \cdot \mathrm{DW}\left(\big[\mathcal{P}_{\mathrm{avg}}; \mathcal{P}_{\max}\big]\right)\big) \\
\mathcal{P}''(X_f) &= \mathcal{P}'(X_f) \cdot R_p
\end{aligned} \tag{10}$$

where $\mathcal{A}_{\mathrm{Res}} = \mathcal{A}''(X_f) - \mathcal{A}(X_f)$, and we choose σ_{p1}, σ_{p3} as the ReLU function, and σ_{p2}, σ_{p4} as Sigmoid function.

Besides, we add an extra post-processing operation to limit the range of phase values to $[-\pi, \pi]$, which is expressed as

$$\mathcal{P}'''(X_f) = \mathcal{P}''(X_f) \cdot 2\pi - \pi. \tag{11}$$

After the recovery in amplitude and phase, we convert them back to real and imaginary parts, expressed as

$$\begin{aligned}
\mathcal{R}'(X_f)(u, v) &= \mathcal{A}''(X_f)(u, v)\ \cos\mathcal{P}'''(X_f)(u, v) \\
\mathcal{I}'(X_f)(u, v) &= \mathcal{A}''(X_f)(u, v)\ \sin\mathcal{P}'''(X_f)(u, v).
\end{aligned} \tag{12}$$

Then, we transform the complex features $\mathcal{F}'(X_f) = \mathcal{R}'(X_f) + j\mathcal{I}'(X_f)$ back to the real features $X_f{}' = \mathrm{ifft}(\mathcal{F}'(X_f))$ by inverse Fourier transform. Finally, we add the original frequency input X'_f to obtain the final frequency branch output features $X''_f = X_f + X'_f$ and concatenate it with the spatial branch output feature X'_s at the end.

To further fuse these two different but complementary types of features, we use an MDTA layer [48] to integrate the features effectively and enhance the representation ability of the model. Given the concatenated input feature $X' = [X'_s; X''_f]$, we first apply a 3×3 convolution layer to get the initial mixed feature $X_n \in \mathbb{R}^{H \times W \times C}$. Then, we follow the work in [48], generate query ($Q = W_d^Q W_p^Q X_n$), key ($K = W_d^K W_p^K X_n$) and value ($V = W_d^V W_p^V X_n$) projections by applying 1×1 convolutions followed by 3×3 depthwise convolutions. Next, we reshape query and key projections such that their dot-product interaction generates a transposed-attention map A of size $\mathbb{R}^{C \times C}$. Overall, the MDTA process is defined as

$$\begin{aligned}
\hat{X}_n &= W_p \mathrm{Attn}(Q, K, V) + X_n \\
\mathrm{Attn}(Q, K, V) &= \mathrm{Softmax}\big(\hat{Q}\hat{K}/\alpha\big) \cdot \hat{V}
\end{aligned} \tag{13}$$

where $\hat{Q} \in \mathbb{R}^{HW \times C}$, $\hat{K} \in \mathbb{R}^{C \times HW}$, and $\hat{V} \in \mathbb{R}^{C \times HW}$ matrices are obtained after reshaping tensors from the original size $\mathbb{R}^{C \times H \times W}$. α is a learnable scaling parameter to control the magnitude of the dot product of $\hat{K}$ and $\hat{Q}$ before applying the softmax function.

C. Cross-Stage Feature Fusion Design

To enhance the ability of TSFNet to propagate useful knowledge from stage one to stage two, we design the novel CFS module. This module adaptively selects the relevant information from both stages and combines them together to generate the final enhanced fusion features. In addition, we employ a GSB [49] as a shortcut connection to alleviate the potential loss of low-frequency information in deeper decoder features. This enhances the representation ability of TSFNet to capture additional clearer frequency information.

As for the process of GSB, given the output features of the latest decoder from stage one, which consists of the spatial branch and frequency branch, denoted as $\hat{X}_n^1 = \{\hat{X}_s^1, \hat{X}_f^1\}$, we first segregate the frequency branch $\hat{X}_f^1$ into multiresolution subtensors. Utilizing the linearity property of the Fourier transform $\mathcal{F}$, we transfer all low-frequency values from larger to smaller tensors, segmenting the frequency features $\hat{X}_f \in \mathbb{C}^{C \times H \times W/2}$ into four levels of Gaussian

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splits $\{\tilde{X}_f^1 \in \mathbb{C}^{C\times H\times(W/2)}, \tilde{X}_f^2 \in \mathbb{C}^{C\times(H/2)\times(W/4)}, \tilde{X}_f^3 \in \mathbb{C}^{C\times(H/4)\times(W/8)}, \tilde{X}_f^4 \in \mathbb{C}^{C\times(H/8)\times(W/16)}\}$, expressed as

$$\tilde{X}_{f\,[i,j]}^1 = \begin{cases} \hat{X}_{f\,[i,j]}^1, & (i,j) \notin \left(\frac{H}{4}\cdots\frac{3H}{4}, 0\cdots\frac{W}{4}\right) \\ \hat{X}_{f\,[i,j]}^1 \cdot \left(1 - \mathcal{N}_{1\,[i,j]}\right) & \\ & (i,j) \in \left(\frac{H}{4}\cdots\frac{3H}{4}, 0\cdots\frac{W}{4}\right) \end{cases} \tag{14}$$

$$\tilde{X}_{f\,[i-\frac{H}{4}, j-\frac{W}{4}]}^2 = \begin{cases} \hat{X}_{f\,[i,j]}^1 \cdot \mathcal{N}_{1\,[i,j]} & \\ \text{when } (i,j) \notin \left(\frac{3H}{8}\cdots\frac{5H}{8}, \frac{3W}{8}\cdots\frac{W}{2}\right) & \\ \hat{X}_{f\,[i,j]}^1 \cdot \mathcal{N}_{1\,[i,j]} \cdot \left(1 - \mathcal{N}_{2\,[i,j]}\right) & \\ \text{when } (i,j) \in \left(\frac{3H}{8}\cdots\frac{5H}{8}, \frac{3W}{8}\cdots\frac{W}{2}\right) & \end{cases} \tag{15}$$

$$\tilde{X}_{f\,[i-\frac{3H}{8}, j-\frac{3W}{8}]}^3 = \begin{cases} \hat{X}_{f\,[i,j]}^1 \cdot \mathcal{N}_{2\,[i,j]} & \\ \text{when } (i,j) \notin \left(\frac{7H}{16}\cdots\frac{9H}{16}, \frac{7W}{16}\cdots\frac{W}{2}\right) & \\ \hat{X}_{f\,[i,j]}^1 \cdot \mathcal{N}_{1\,[i,j]} \cdot \mathcal{N}_{2\,[i,j]} \cdot \left(1 - \mathcal{N}_{3\,[i,j]}\right) & \\ \text{when } (i,j) \in \left(\frac{7H}{16}\cdots\frac{9H}{16}, \frac{7W}{16}\cdots\frac{W}{2}\right) & \end{cases} \tag{16}$$

$$\tilde{X}_f^4 = \hat{X}_{f\,[\frac{7H}{16}\cdots\frac{9H}{16}, \frac{7W}{16}\cdots\frac{W}{2}]}^1 \cdot \mathcal{N}_1 \cdot \mathcal{N}_2 \cdot \mathcal{N}_3 \tag{17}$$

where $\mathcal{N}_1$, $\mathcal{N}_2$, and $\mathcal{N}_3$ are Gaussian weight maps like in [49], the centers of them are all at $((H/2), 0)$ and the standard deviations are proportional to the corresponding resolution.

Subsequently, we sum these Gaussian split frequency features with the original decoder frequency features from stage one, resulting in enhanced decoder features $\hat{X}_n^i = \{\hat{X}_s^i, \hat{X}_f^i + \tilde{X}_f^i\}$.

After that, the CFS module adaptively fuses these enhanced stage one decoder features $\hat{X}_n^i$ and the corresponding stage two encoder features $\hat{X}_n^{i'}$. Initially, as shown in Fig. 4, we use two 3 × 3 convolution layers to process features from both stages, followed by concatenation $\tilde{X}_n^i = \{\text{Conv}_{3\times3}(\hat{X}_n^i), \text{Conv}_{3\times3}(\hat{X}_n^{i'})\}$. Then, we employ channel-based average and maximum pooling, denoted as $P_{\text{avg}}(\cdot)$ and $P_{\text{max}}(\cdot)$, to squeeze the channel information into two representative feature maps, expressed as

$$\begin{aligned} X_{\text{avg}} &= P_{\text{avg}}\left(\tilde{X}_n^i\right) \\ X_{\text{max}} &= P_{\text{max}}\left(\tilde{X}_n^i\right). \end{aligned} \tag{18}$$

To further enable the spatial information interaction among two stage features, we employ a large kernel $\text{Conv}_{7\times7}(\cdot)$ convolution layer with Sigmoid activation to select the important features region by region and generate the spatial selection maps SA that needs to be enhanced and propagated, expressed as

$$SA = \sigma\left(\text{Conv}_{7\times7}\left(\left[X_{\text{avg}}; X_{\text{max}}\right]\right)\right). \tag{19}$$

Finally, the CFS module outputs are obtained by applying a 3 × 3 convolution layer to the sum of the elementwise products between the input stage features and their respective spatial selection maps, expressed as

$$\hat{X}_n^{i''} = \text{Conv}_{3\times3}\left(\hat{X}_n^i \cdot SA_1 + \hat{X}_n^{i'} \cdot SA_2\right). \tag{20}$$

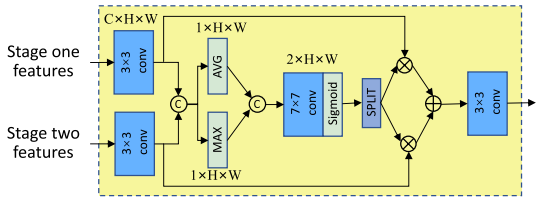


Fig. 4. Illustration of the CFS module.

IV. Experimental Results and Analyses

A. Experiment Setup

1) Datasets and Configuration: This study evaluates the effectiveness of our TSFNet on three public remote sensing datasets: UCMerced [52], NWPU-RESISC45 [53], and Aerial Image dataset (AID) [54].

1) **UCMerced dataset [52]** comprises 21 remote sensing scenes, each containing 100 images with 256 × 256 resolution. To train and test our model, we adopt the dataset split methodology in [8] and [13], and divide the data set into two equal parts as the training set and the test set. In addition, 10% in the training set is used to conduct the val set.
2) **NWPU-RESISC45 dataset [53]** consists of 45 different scenes, with 700 images per scene. For our experiments, we randomly selected 100 images from each scene type. Similar to the above methodology, we split the randomly selected dataset into two balanced halves for training and testing, with 50 images in each scene. A validation set is also created using 10% of the training set.
3) **AID [54] dataset** comprises 10 000 images with 600 × 600 resolution in 30 classes of remote sensing scenes. Following the approach in [13], we randomly allocate 80% of the dataset for training, while reserving five images per class for validation. The remaining 20% of the images are utilized as the test set.

In our experiment, we concentrate on large-factor RSISR with scaling factors of ×8 and ×16. During the training phase, we randomly crop 256 × 256 HR image patches from the training set as ground truth. The corresponding LR inputs are generated by downsampling these patches through bicubic interpolation. We evaluate the quantitative performance using the traditional peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) [55]. Higher PSNR and SSIM values indicate better-reconstructed image quality. Besides, we also evaluate the comparison RSISR results with

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a scaling factor of $\times 4$ in three datasets to further validate the effectiveness of our method.

2) Implementation Details: The number of channel dimensions in our TSFNet is set to 64. For optimization, we employ the Adam optimizer with parameters $\beta_1 = 0.9$, $\beta_2 = 0.99$, and $\varepsilon = 10^{-8}$. The initial learning rate is set to $3e^{-4}$, and we use a mini-batch size of 4. The initial learning rate is 3e-4 and reduces by 50% in 300, 700, 900, and 1300 epochs for a total of 1500 epochs, respectively.

B. Comparisons With State-of-the-Art Methods

To evaluate the effectiveness of our proposed method, we compare the proposed method with some leading natural image SR methods, including D-DBPN [29], SwinIR [32], U-shaped transformer network (Uformer) [56], super-resolution network with a deep dictionary (SRDD) [57], HATNet [33], and permuted self-attention for single image super-desolution (SRFormer) [58], and some RSISR methods, including deep compendium model (DCM) [12], mixed high-order attention network (MHAN) [42], HSENet [8], TransENet [13], MAGAN [2], and SpecGAN [3]. Notably, we exclude the Registration and fusion block for the HSIs input when evaluating the SpecGAN method, as it is specifically designed for additional HSIs-guided SR of RSIs, ensuring a fair comparison.

Table I presents quantitative analyses using different objective evaluation metrics on three datasets. Our proposed TSFNet consistently achieves the best performance results on both metrics for the smaller UCMerced and NWPU-RESISC45 datasets. Meanwhile, TSFNet also exhibits competitive performance on the larger AID datasets, outperforming other methods. Table II compares the model latency and performance accuracy of the mentioned methods with our TSFNet. This comparison results reveal that our TSFNet achieves a better balance between parameters and accuracy, despite its increased computational consumption and longer inference time. This additional time and computational cost are attributed to the preupsample operation, which is essential for recovering the base detail in the frequency domain and preventing the blurring or confusing extraction of frequency features. Nevertheless, our inference time and floating point operations per second (FLOPs) remain lower than those of the D-DBPN [29] and SpecGAN [3] methods, despite having a significantly lower parameter count. Given that the large-factor RSISR task presents a more challenging scenario, we consider this cost to be acceptable and effective for the large-factor RSISR task.

Furthermore, we present a qualitative comparison of the SR results obtained using various networks on three test datasets, as depicted in Fig. 5. On the UCMerced dataset, compared with the HR ground truths, SR results from other CNN-based (DCM, MHAN, HSENet, and SRDD), transformer-based methods (TransENet, Uformer, HATNet, and SRformer), or GAN-based methods (MAGAN and SpecGAN) all exhibit severe deviations in capturing the global structural shape in the freeway scenario. Specifically, in the freeway scenario, they cannot accurately recover the shapes of both upper and lower roads in all regions, even when some of them possess strong global similarity modeling capabilities. In contrast, our

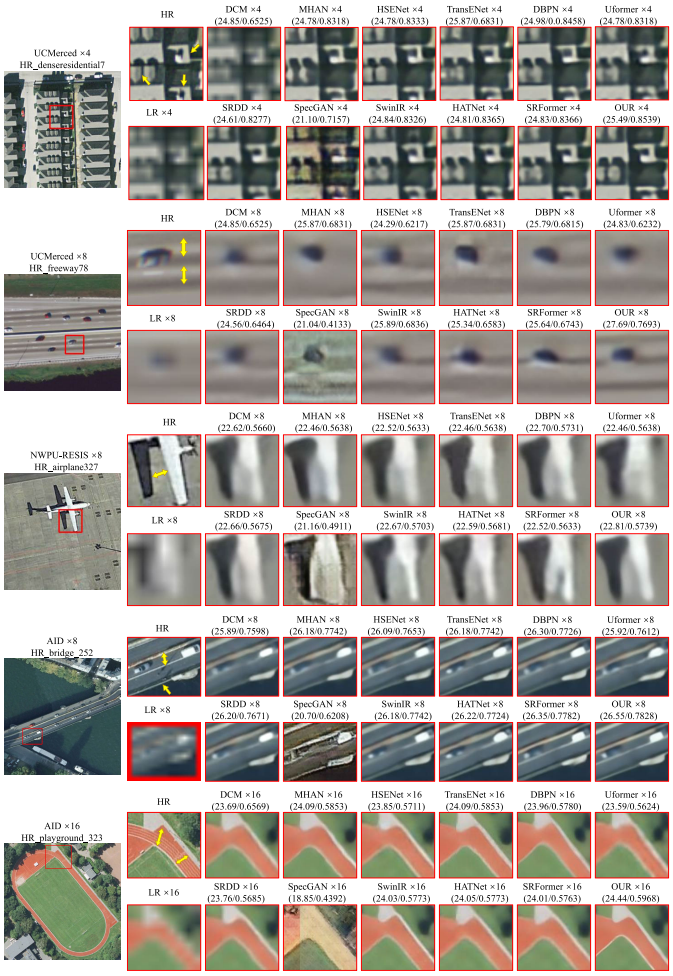


Fig. 5. Result comparisons on different datasets with different methods. Please zoom in for the best view.

TSFNet performs much better recovery on the global road shape. This indicates that our method is indeed capable of effectively extracting global regularities of road information, rather than solely relying on similarity judgments. On the NWPU-RESISC45 dataset, transformer-based methods, such as TransENet and SRFormer, exhibit obvious ghost phenomena at the boundary between the shadow and wing parts, resulting from the erroneous fusion of visually similar but semantically unrelated regions. Our TSFNet effectively alleviates this phenomenon and generates clearer visual results. Moreover, on the bigger AID dataset, our method consistently preserves clearer edge contours in the playground and bridge scenarios. In addition, following [8] and [43], we select real-world RSIs from GaoFen-1 and GaoFen-2 to demonstrate the practical reconstructive capabilities of our method, as depicted in Fig. 6. Specifically, we use the pretrained models on the AID dataset to reconstruct the details of real images. Despite variances between these images and the training dataset, our TSFNet still excels in capturing additional details, particularly global rich shape information. This observation further demonstrates the effectiveness of our approach.

C. Ablation Studies

All ablation studies are performed on the UCMerced dataset for $\times 8$ SR.

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TABLE I

MEAN PSNR (dB) AND SSIM OVER THREE DIFFERENT TEST DATASETS. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY. PLEASE ZOOM IN FOR THE BEST VIEW

Dataset	Metric	Bicubic	D-DBPN [29]	DCM [12]	MHAN [42]	SwinIR [32]	HSENet [8]	TransE-Net [13]	Uformer [56]	MAGAN [2]	SRDD [57]	SpecGAN [3]	HATNet [33]	SRFormer [58]	TSFNet (Ours)
UCMerced ×4	PSNR	25.6	27.7	27.53	27.56	27.73	27.68	27.70	27.24	25.63	27.6	22.51	27.6	27.82	27.92
	SSIM	0.6729	0.7627	0.7556	0.7581	0.7643	0.7611	0.7635	0.7436	0.6860	0.7609	0.6327	0.7591	0.7685	0.7695
UCMerced ×8	PSNR	22.55	23.70	23.59	23.49	23.69	23.56	23.56	23.43	22.89	23.53	19.94	23.66	23.69	23.90
	SSIM	0.5059	0.5727	0.5619	0.5608	0.5759	0.5665	0.5654	0.5513	0.5315	0.5623	0.4129	0.5732	0.5759	0.5852
UCMerced ×16	PSNR	20.57	21.11	21.04	20.97	20.79	21.05	20.98	20.98	20.45	20.98	18.17	21.03	21.05	21.20
	SSIM	0.4358	0.4588	0.4510	0.4561	0.4448	0.4515	0.453	0.4457	0.4314	0.4498	0.3732	0.4540	0.4557	0.4635
NWPU RESISC45 ×4	PSNR	25.12	25.91	25.86	25.68	25.92	25.91	25.91	25.79	25.12	25.82	22.39	25.94	25.96	26.01
	SSIM	0.5998	0.6458	0.6433	0.6350	0.6468	0.6471	0.6456	0.6396	0.6075	0.6443	0.5397	0.6472	0.6486	0.6516
NWPU RESISC45 ×8	PSNR	22.82	23.23	23.23	23.24	23.25	23.24	23.21	23.20	23.08	23.17	20.04	23.23	23.24	23.29
	SSIM	0.4825	0.5099	0.5085	0.5100	0.5100	0.5087	0.5075	0.5059	0.4946	0.5080	0.3632	0.5090	0.5102	0.5124
NWPU RESISC45 ×16	PSNR	21.11	21.31	21.32	21.30	21.18	21.31	21.32	21.31	21.14	21.28	18.78	21.28	21.31	21.35
	SSIM	0.4347	0.4449	0.4444	0.4451	0.4396	0.4439	0.4444	0.4428	0.4146	0.4425	0.3016	0.4440	0.4447	0.4465
AID ×4	PSNR	27.30	29.29	29.20	29.39	29.47	29.32	29.44	29.25	26.99	29.21	23.81	29.49	29.42	29.50
	SSIM	0.7039	0.7856	0.7826	0.7892	0.7923	0.7867	0.7912	0.7850	0.7134	0.7835	0.6913	0.7934	0.7918	0.7926
AID ×8	PSNR	24.39	25.67	25.50	25.56	25.71	25.61	25.71	25.54	24.22	25.53	21.21	25.78	25.75	25.77
	SSIM	0.5604	0.6248	0.6155	0.6193	0.6291	0.6213	0.6293	0.6186	0.5577	0.6176	0.4895	0.6309	0.6312	0.6315
AID ×16	PSNR	22.47	23.27	23.15	23.17	23.27	23.22	23.19	23.16	22.47	23.18	19.61	23.29	23.29	23.36
	SSIM	0.5026	0.5298	0.5234	0.5263	0.5306	0.5264	0.5288	0.5235	0.4931	0.5248	0.3887	0.5312	0.5305	0.5346

TABLE II

QUANTITATIVE COMPARISON OF PARAMETERS, FLOPS, AND PSNR ON UCMERCED DATASET FOR ×8 SR

Method	Pub.	Param (M)	FLOPs (G)	Time (ms)	PSNR (dB)
D-DBPN [29]	CVPR 2018	23.20	2370.75	169.19	23.70
DCM [12]	IEEE GRSL 2019	2.51	8.70	2.41	23.59
MHAN [42]	IEEE TGRS 2020	11.50	16.97	14.28	23.49
SwinIR [32]	ICCV 2021	12.00	15.14	33.00	23.69
HSENet [8]	IEEE TGRS 2021	5.58	7.30	30.90	23.56
TransENet [13]	IEEE TGRS 2021	37.61	17.01	28.66	23.56
MAGAN [2]	IEEE TGRS 2022	0.65	7.46	6.83	22.89
Uformer [56]	CVPR 2022	20.71	0.75	23.44	23.43
SRDD [57]	ECCV 2022	10.35	1.65	35.14	23.53
SpecGAN [3]	IEEE TGRS 2022	139.12	131.71	137.11	19.94
HATNet [33]	CVPR 2023	20.66	24.10	72.30	23.66
SRFormer [58]	ICCV 2023	10.58	30.89	54.22	23.69
TSFNet	–	3.13	55.59	136.70	23.90

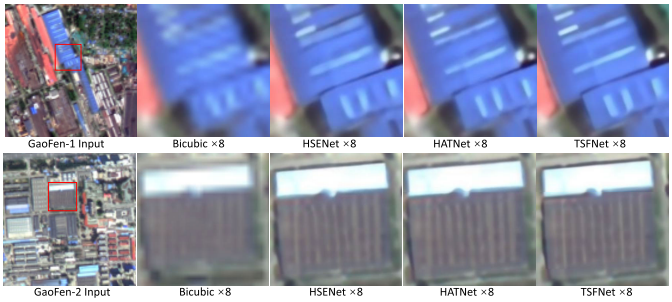


Fig. 6. Result comparisons on real GaoFen-1 and GaoFen-2 satellite data. Please zoom in for the best view.

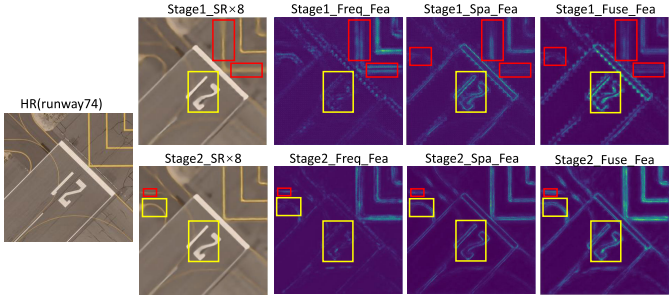


Fig. 7. SR results and visualization of the feature fusion process between spatial domain and frequency domain.

1) Effects of Dual-Branch SFE Design: The SFE modules play a crucial role in our proposed network. Table III lists quantitative comparison results on the UCMerced ×8 dataset under different settings. Compared to configurations using only residual blocks, our SFE can effectively reduce the number of parameters by more than half while maintaining the same PSNR accuracy in the single-stage setup and even improving the PSNR accuracy by 0.03 dB in the two-stage setup. This indicates the effectiveness of our AGPF block in the SFE module. Besides, we also investigate the influence of MDTA layer [48] used in our TSFNet. It can achieve an improvement of 0.06 dB in the single-stage setup and 0.04 dB in the two-stage setup, which proves its effectiveness in our

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TABLE III
ABLATION EXPERIMENTS FOR THE BLOCKS ON UCMERCED-×8

One-Stage	RSPB	SFE	MDTA	Two-Stage	GSB	CFS	Param (M)	PSNR (dB)	SSIM
✓	✓						2.79	23.73	0.5748
✓		✓					1.05	23.73	0.5755
✓		✓	✓				1.49	23.79	0.5813
	✓			✓			5.59	23.76	0.5789
		✓		✓			2.09	23.79	0.5802
		✓	✓	✓			2.97	23.83	0.5849
		✓	✓	✓	✓		2.97	23.88	0.5851
		✓	✓	✓		✓	3.13	23.88	0.5844
		✓	✓	✓	✓	✓	3.13	23.90	0.5852

TABLE IV
ABLATION EXPERIMENTS FOR THE USAGE OF TWO-STAGE AND CFS STRATEGIES ON UCMERCED-×8

Basiline	Two-Stage	CFS	Param (M)	PSNR (dB)	SSIM
SwinIR [32]			12.00	23.69	0.5759
	✓		24.28	23.71	0.5730
	✓	✓	24.88	23.74	0.5733
Uformer [56]			20.71	23.43	0.5513
	✓		41.44	23.49	0.5555
	✓	✓	42.23	23.53	0.5572
SRFormer [58]			10.57	23.67	0.5750
	✓		21.44	23.68	0.5746
	✓	✓	22.03	23.70	0.5765
TSFNet			1.49	23.79	0.5813
	✓		2.97	23.83	0.5849
	✓	✓	3.13	23.88	0.5844

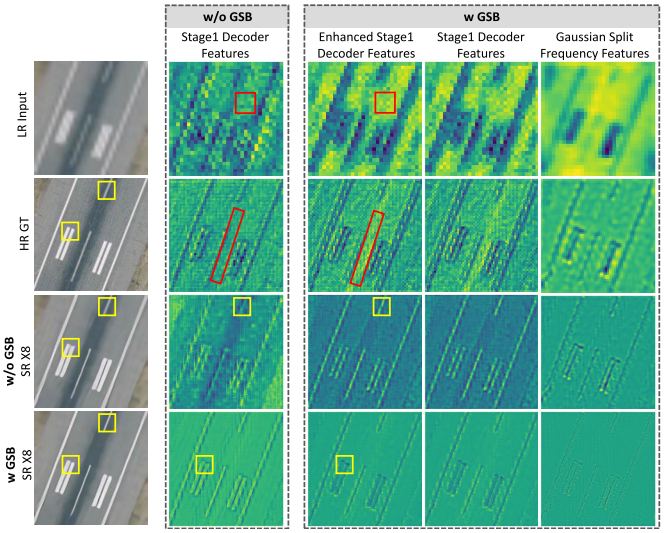


Fig. 8. Feature visualization of four-level decoders in stage one and SR results with GSB and without GSB. The second column of images depicts the outputs of four-level decoder features in stage one under the ablation experiment without the GSB setting. The third column of images represents the outputs of the enhanced decoder features after GSB in stage one, while the fourth column images depict the outputs of decoder features before GSB and the fifth column images represents the Gaussian split frequency features of GSB. The red boxes highlight the Gaussian split multiresolution frequency features added by GSB, while the yellow boxes compare the locations where features are generated with and without fidelity due to the GSB.

TSFNet by integrating the features from frequency and spatial domains.

To delve deeper into our dual domain information exploration mechanism, we perform an in-depth analysis of the final layer feature maps from both stages of TSFNet. Through qualitative visual comparisons, as illustrated in Fig. 7, we highlight the complementary nature of frequency features extracted by AGPF and spatial features extracted by RSPB. For example, the red boxes in Fig. 7 highlight the presence of finer frequency features, while the yellow boxes emphasize detailed spatial features. Moreover, the MDTA fusion module effectively fuses the advantages of both feature types. By integrating them, the response of the fused features becomes more informative, showcasing the powerful capability of our core module.

2) Effects of Two-Stage Architecture Design and Information Propagation Between Two Stages: In Table III, we delve into the impact of our two-stage architecture design, alongside GSB and CFS module, on the propagation of information between the two stages. We notice that compared to the single-stage U-Net architecture, the two-stage architecture can obtain a 0.03 dB improvement with only residual blocks, and this enhancement is further elevated to 0.06 dB using SFE blocks. This indicates the effectiveness of our two-stage architecture design. Regarding the GSB, our analysis shows that it contributes to an increase of 0.05 dB in PSNR without increasing the parameters. As for the CFS module, we can see that our CFS module also results in a 0.05 dB improvement in PSNR. To further demonstrate the practical reconstructive capabilities of the two-stage refined architecture and CFS-designed adaptive information enhancement, we integrate these elements into existing state-of-the-art networks such as SwinIR [32], Uformer [56], and SRFormer [58]. The outcomes of this integration are summarized in Table IV. It is evident that the two-stage architecture doubles the size of

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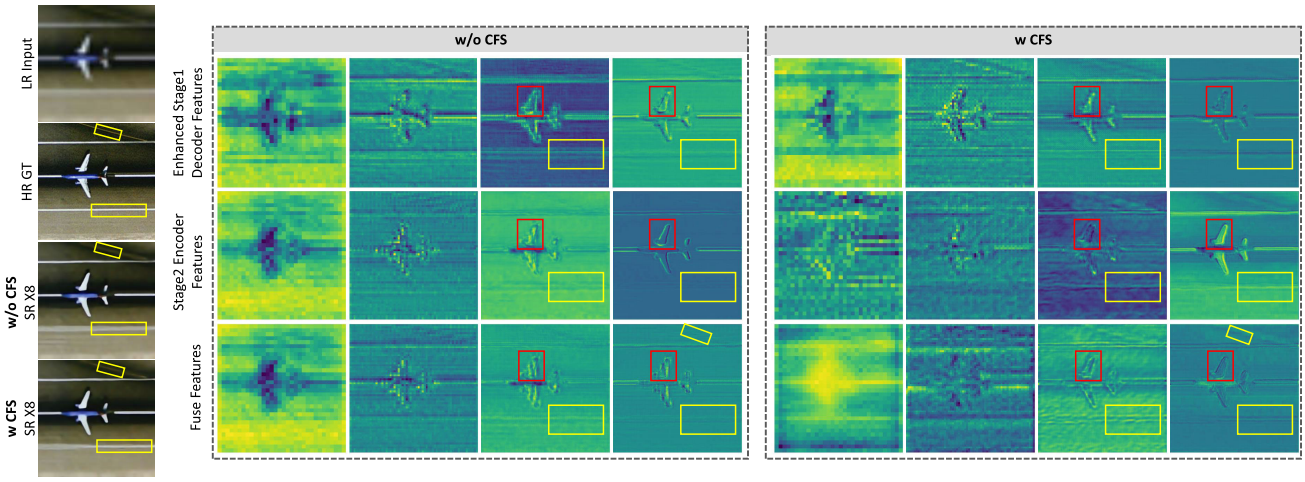


Fig. 9. Feature visualization of four-level encoders in stage two and SR results with CFS and without CFS. The red boxes emphasize the enhanced adaptive fusion ability facilitated by CFS. The yellow boxes compare the feature generation locations with and without fidelity, attributed to the presence of CFS.

model parameters and improves the reconstruction accuracy across all network baselines. These improvements validate the effectiveness of our design that the progressive recovery from coarse to fine proves to be beneficial for the large-factor RSISR task. In addition, the CFS design in our article further improves model accuracy under the two-stage architecture across all network baselines with minimal parameter cost, demonstrating its practical and promising application in the large-factor RSISR task.

To gain a more intuitive understanding of the information propagation process brought by the GSB and CFS modules within the two-stage architecture, we visualize the corresponding features affected by these modules in Figs. 8 and 9. In Fig. 8, the comparison of feature visualizations between the second and third columns indicates that GSB is helpful to provide more enhanced feature details, leading to improved fidelity in the SR results. Specifically, the multiresolution Gaussian split features (shown in the fifth column) exhibit higher intensity in the flat lower frequency region compared to the edge higher frequency part, demonstrating GSB's effectiveness in conveying lower frequency information in the lower abstract decoder levels. In Fig. 9, the comparison of the fusion results between the enhanced decoder features from stage one and the encoder features from stage two reveals that our CFS design offers greater flexibility in the adaptive fusion of two-stage features compared to the ablation setting using simple summation without CFS. Specifically, when fusing the last two higher resolution features, our CFS focuses on capturing finer edges and better texture information from both stages, while the simple summation fusion results confuse the difference between the two stages and cannot fuse effectively from two stages. In addition, when the CFS works on the first two lower resolution features, it exhibits a more abstract behavior than the summation fusion, but our CFS places a larger emphasis on capturing global abstract shape information compared to the summation results which are higher in the flat region than in the edge part. Finally, the first column SR results present that our CFS achieves competitive visual outcomes and further demonstrate the advancement of our CFS.

V. Conclusion

In this article, we propose TSFNet, a novel TSFNet for large-factor RSISR. Through the AGPF block design, TSFNet can explicitly utilize the meaningful physical disentangling prior characteristics in the Fourier transformation and recover the global common and specific structural features in the frequency domain effectively. In addition, we introduce a two-stage progressive feature learning architecture to refine the results from coarse to fine. The cross-stage feature fusion design facilitates the propagation of useful frequency information and knowledge from stage one to stage two. The experimental results clearly demonstrate the superior performance of TSFNet compared to other state-of-the-art techniques in large-factor RSISR, highlighting the effectiveness of our method.

However, it is important to acknowledge that our current research is primarily focused on large-factor SR tasks under the bicubic degradation model, utilizing small-volume optical RSI datasets. Consequently, while our method can partially address the issue of reconstruction blur, it faces challenges in handling a variety of complex and unknown degradation scenarios encountered in real-world applications. In our future work, we aim to further extend our exploration to the impact of frequency recovery on these challenging real-world degradation problems. In addition, we intend to delve into the realm of large-factor HSI SR, integrating additional frequency information and higher order spectral information to assist in this challenging task.

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